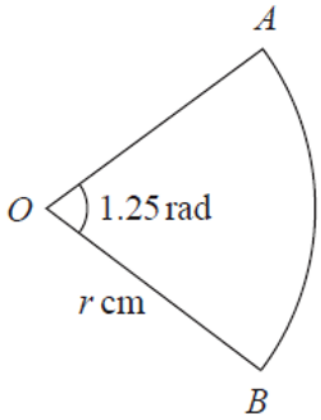
Questions

Q1.



RocketRevise

Figure 1

Figure 1 shows a sector AOB of a circle with centre O and radius r cm.

The angle AOB is 1.25 radians.

Given that the area of the sector AOB is 15 cm^2

(a) find the exact value of r,

(2)

(b) find the exact length of the perimeter of the sector. Write your answer in simplest form.

(3)

(Total for question = 5 marks)

www.RocketRevise.com

(Q01 WMA11/01, Oct 2019)

Q2.

compiled by Ahmed Gul www.RocketRevise.com

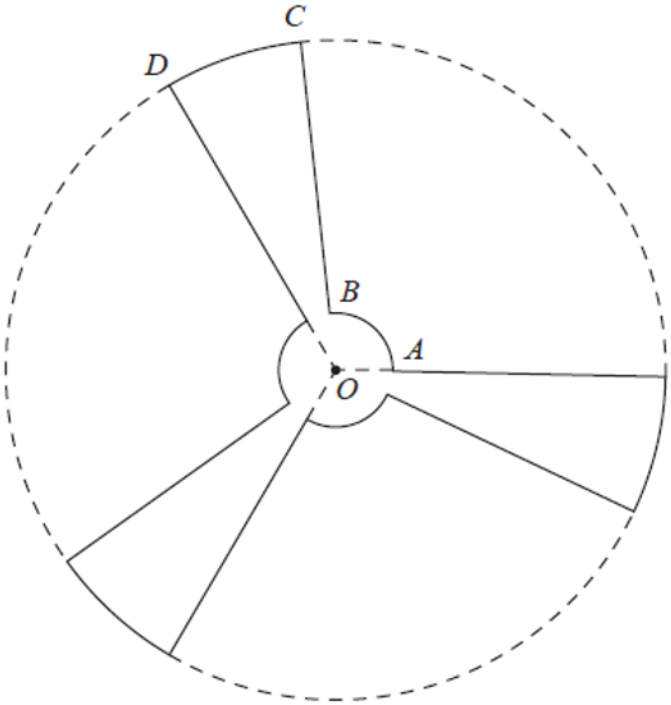


Figure 3

RocketRevise

Figure 3 shows a sketch of the outline of the face of a ceiling fan viewed from below.

The fan consists of three identical sections congruent to *OABCDO*, shown in Figure 3, where

- *OABO* is a sector of a circle with centre *O* and radius 9 cm
- *OBCDO* is a sector of a circle with centre *O* and radius 84 cm
- angle $AOD = \frac{2\pi}{3}$ radians

Given that the length of the arc *AB* is 15 cm,

(a) show that the length of the arc *CD* is 35.9 cm to one decimal place.

(3)

The face of the fan is modelled to be a flat surface.

Find, according to the model,

(b) the perimeter of the face of the fan, giving your answer to the nearest cm,

www.RocketRevise.com

(2)

(c) the surface area of the face of the fan.

Give your answer to 3 significant figures and make your units clear.

(5)

(Total for question = 10 marks)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(**Q08 WMA11/01, June 2022**)

Q3.

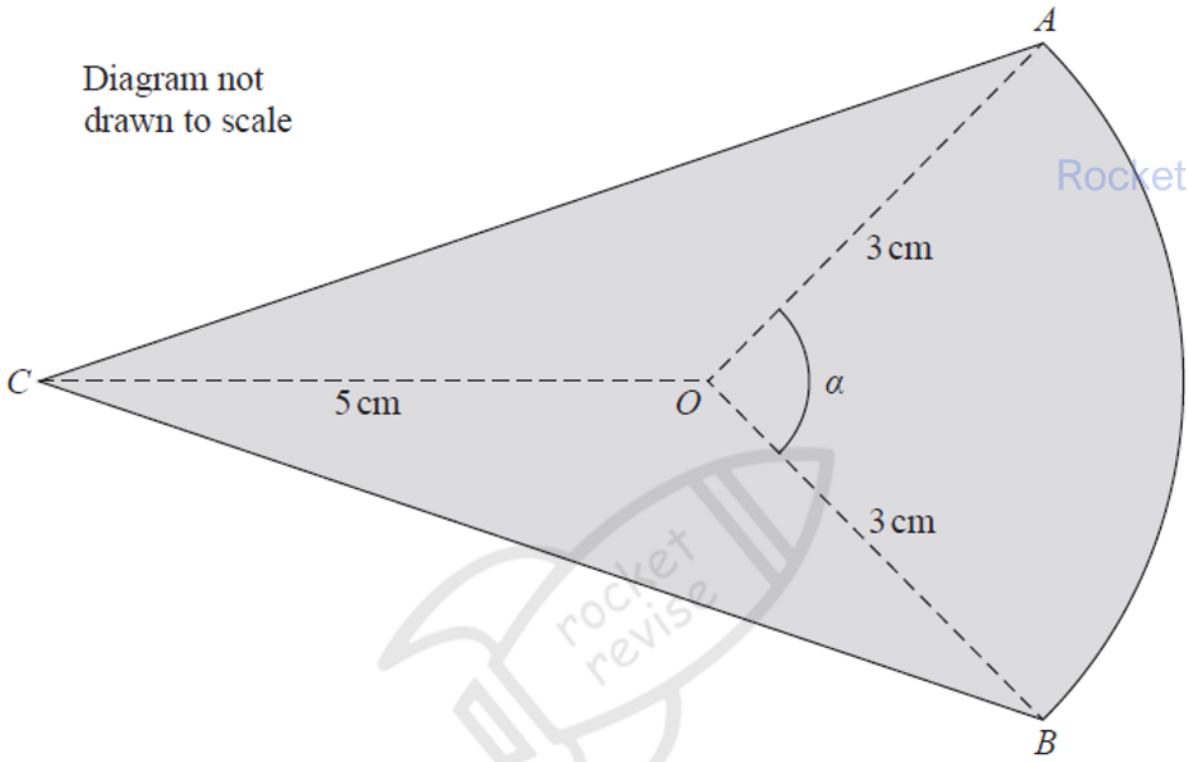


Figure 1

Figure 1 shows the design for a badge.

The design consists of two congruent triangles, *AOC* and *BOC*, joined to a sector *AOB* of a circle centre *O*.

- Angle *AOB* = α
- *AO* = *OB* = 3 cm
- *OC* = 5 cm

www.RocketRevise.com

Given that the area of sector *AOB* is 7.2 cm^2

(a) show that α = 1.6 radians.

(2)

(b) Hence find

(i) the area of the badge, giving your answer in cm^2 to 2 significant figures,

(ii) the perimeter of the badge, giving your answer in cm to one decimal place.

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(8)

(Total for question = 10 marks)

(Q03 WMA11/01, Oct 2020)

Q4.

RocketRevise

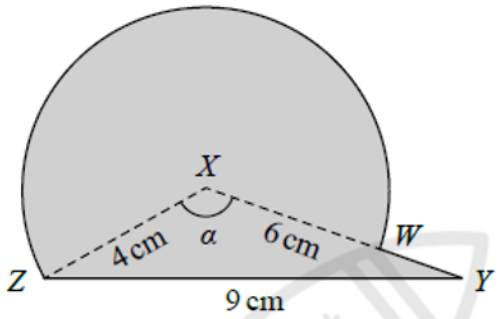


Figure 4

The triangle XYZ in Figure 4 has $XY = 6$ cm, $= 9$ cm, $ZX = 4$ cm and angle $ZXY = \alpha$.

The point W lies on the line XY.

The circular arc ZW, in Figure 4, is a major arc of the circle with centre X and radius 4 cm.

(a) Show that, to 3 significant figures, $\alpha = 2.22$ radians.

(2)

(b) Find the area, in cm^2, of the major sector $XZWX$.

(3)

The region, shown shaded in Figure 4, is to be used as a design for a logo.

www.RocketRevise.com

Calculate

(c) the area of the logo

(3)

(d) the perimeter of the logo.

(4)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(Total for question = 12 marks)

(Q10 WMA11/01, Specimen papers)

Q5.

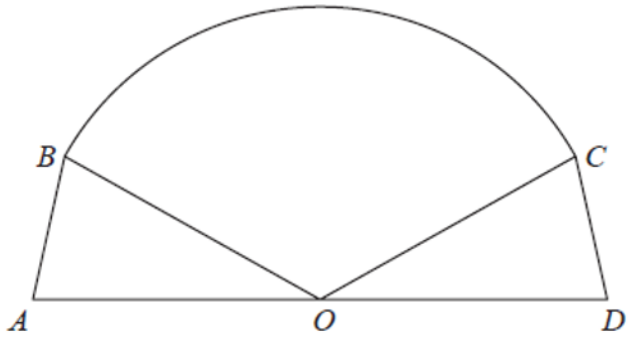


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the plan view for the design of a stage.

The design consists of a sector *OBC* of a circle, with centre *O*, joined to two congruent triangles *OAB* and *ODC*.

Given that

- angle *BOC* = 2.4 radians
- area of sector *BOC* = 40 m^2
- *AOD* is a straight line of length 12.5 m

(a) find the radius of the sector, giving your answer, in m, to 2 decimal places,

(2)

(b) find the size of angle *AOB*, in radians, to 2 decimal places.

www.RocketRevise.com

(1)

Hence find

(c) the total area of the stage, giving your answer, in m^2, to one decimal place,

(3)

(d) the total perimeter of the stage, giving your answer, in m, to one decimal place.

(4)

www.RocketRevise.com

compiled by Ahmed Gul
www.RocketRevise.com

(Total for question = 10 marks)

(Q06 WMA11/01, Jan 2023)

Q6.

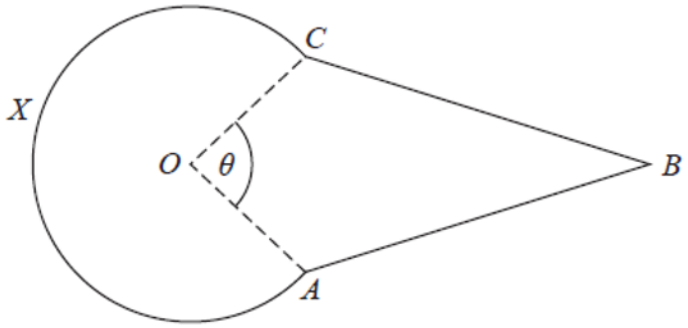


Figure 3

Figure 3 shows the design for a sign at a bird sanctuary.

The design consists of a kite *OABC* joined to a sector *OCXA* of a circle centre *O*.

In the design

- *OA* = *OC* = 0.6 m
- *AB* = *CB* = 1.4 m
- Angle *OAB* = Angle *OCB* = 2 radians
- Angle *AOC* = θ radians, as shown in Figure 3

Making your method clear,

(a) show that $\theta = 1.64$ radians to 3 significant figures,

(4)

(b) find the perimeter of the sign, in metres to 2 significant figures,

(2)

(c) find the area of the sign, in m^2 to 2 significant figures.

(4)

(Total for question = 10 marks)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(**Q07 WMA11/01, Oct 2021**)

Q7.

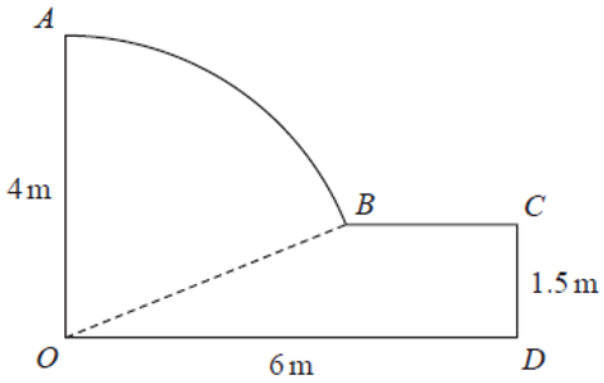


Figure 1

Figure 1 shows the plan for a garden.

In the plan

- *OA* and *CD* are perpendicular to *OD*
- *AB* is an arc of the circle with centre *O* and radius 4 metres
- *BC* is parallel to *OD*
- *OD* is 6 metres, *OA* is 4 metres and *CD* is 1.5 metres

(a) Show that angle *AOB* is 1.186 radians to 4 significant figures.

(2)

(b) Find the perimeter of the garden, giving your answer in metres to 3 significant figures.

(4)

www.RocketRevise.com

(c) Find the area of the garden, giving your answer in square metres to 3 significant figures.

(4)

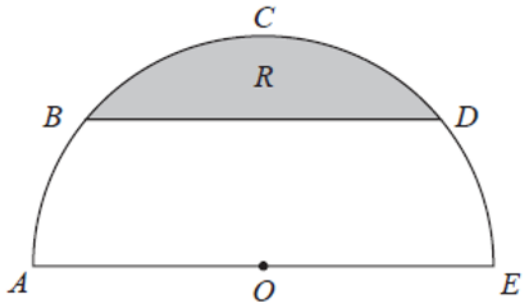
(Total for question = 10 marks)

(**Q05 WMA11/01, June 2023**)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

Q8.



Not to scale

Figure 2

RocketRevise

Figure 2 shows a plan view of a semicircular garden *ABCDEOA*

The semicircle has

- centre *O*
- diameter *AOE*
- radius 3 m

The straight line *BD* is parallel to *AE* and angle *BOA* is 0.7 radians.

(a) Show that, to 4 significant figures, angle *BOD* is 1.742 radians.

(1)

The flowerbed *R*, shown shaded in Figure 2, is bounded by *BD* and the arc *BCD*.

(b) Find the area of the flowerbed, giving your answer in square metres to one decimal place.

(3)

(c) Find the perimeter of the flowerbed, giving your answer in metres to one decimal place.

(3)

www.RocketRevise.com

(Total for question = 7 marks)

(Q05 WMA11/01, Jan 2022)

Q9.

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

A sector AOB, of a circle centre O, has radius r cm and angle θ radians.

Given that the area of the sector is 6 cm^2 and that the perimeter of the sector is 10 cm,

(a) show that

$$3\theta^2 - 13\theta + 12 = 0$$

(4)

(b) Hence find possible values of r and θ.

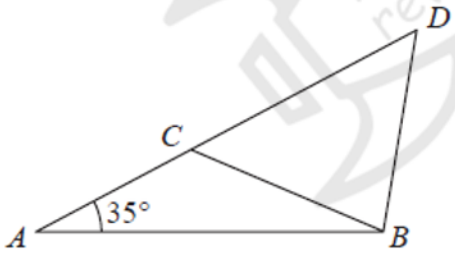
(3)

RocketRevise

(Total for question = 7 marks)

(Q10 WMA11/01, Jan 2019)

Q10.



Not to scale

Figure 3

Figure 3 shows the design for a structure used to support a roof.

The structure consists of four wooden beams, AB, BD, BC and AD.

Given $AB = 6.5\ m$, $BC = BD = 4.7\ m$ and angle $BAC = 35°$

www.RocketRevise.com

(a) find, to one decimal place, the size of angle ACB,

(3)

(b) find, to the nearest metre, the total length of wood required to make this structure.

(3)

(Total for question = 6 marks)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(**Q07 WMA11/01, Jan 2019**)

Q11.

A parallelogram *ABCD* has area 40 cm^2

Given that *AB* has length 10 cm, *BC* has length 6 cm and angle *DAB* is obtuse, find

(a) the size of angle *DAB*, in degrees, to 2 decimal places,

RocketRevise

(3)

(b) the length of diagonal *BD*, in cm, to one decimal place.

(2)

(Total for question = 5 marks)

(**Q04 WMA11/01, Oct 2019**)

Q12.

In the triangle *ABC*,

- *AB* = 21 cm
- *BC* = 13 cm
- angle *BAC* = 25°
- angle *ACB* = *x*°

www.RocketRevise.com

(a) Use the sine rule to find the value of sin *x*° , giving your answer to 4 decimal places.

(2)

Given also that *AB* is the longest side of the triangle,

(b) find the value of *x*, giving your answer to 2 decimal places.

(3)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(Total for question = 5 marks)

(Q02 WMA11/01, June 2022)

Q13.

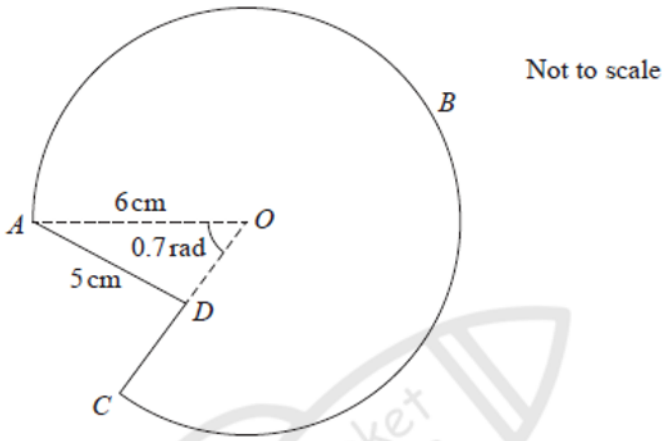


Figure 2

The shape *ABCDA* consists of a sector *ABCOA* of a circle, centre *O*, joined to a triangle *AOD*, as shown in Figure 2.

The point *D* lies on *OC*.

The radius of the circle is 6 cm, length *AD* is 5 cm and angle *AOD* is 0.7 radians.

(a) Find the area of the sector *ABCOA*, giving your answer to one decimal place.

(3)

Given angle *ADO* is obtuse,

(b) find the size of angle *ADO*, giving your answer to 3 decimal places.

(3)

(c) Hence find the perimeter of shape *ABCDA*, giving your answer to one decimal place.

(4)

(Total for question = 10 marks)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

Q14.

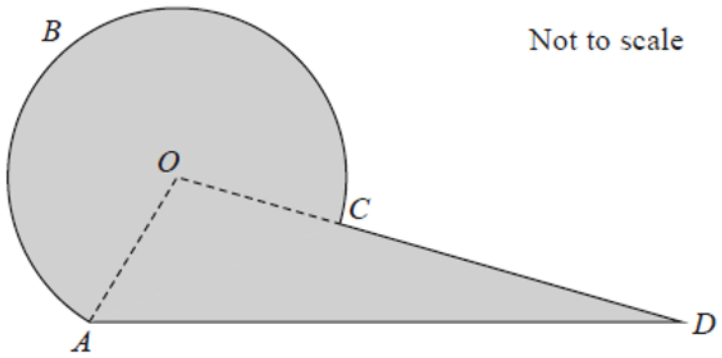


Figure 3

Figure 3 shows the plan view of a viewing platform at a tourist site.

The shape of the viewing platform consists of a sector *ABCOA* of a circle, centre *O*, joined to a triangle *AOD*.

Given that

- *OA* = *OC* = 6 m
- *AD* = 14 m
- angle *ADC* = 0.43 radians
- angle *AOD* is an obtuse angle
- *OCD* is a straight line

find

(a) the size of angle *AOD*, in radians, to 3 decimal places,

(3)

(b) the length of arc *ABC*, in metres, to one decimal place,

(2)

(c) the total area of the viewing platform, in m^2, to one decimal place.

(4)

(Total for question = 9 marks)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

Q15.

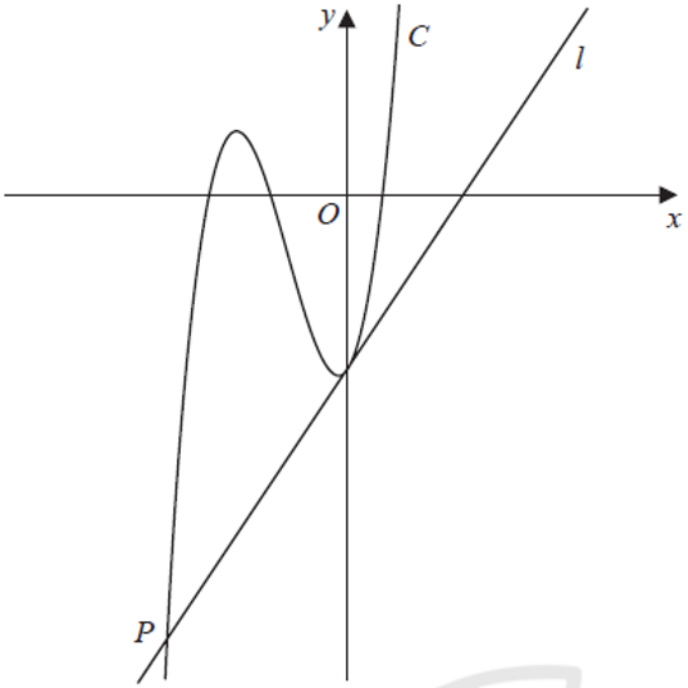


Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x + 20)(x + 6)(2x - 3)$$

(a) Use the given information to state the values of x for which

$$f(x) > 0$$

(2)

(b) Expand $(3x + 20)(x + 6)(2x - 3)$, writing your answer as a polynomial in simplest form.

(3)

The straight line l is the tangent to C at the point where Ccuts the y-axis.

Given that l cuts C at the point P, as shown in Figure 4,

www.RocketRevise.com

(c) find, using algebra, the x coordinate of P

(*Solutions based on calculator technology are not acceptable.*)

(5)

(Total for question = 10 marks)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

Q16.

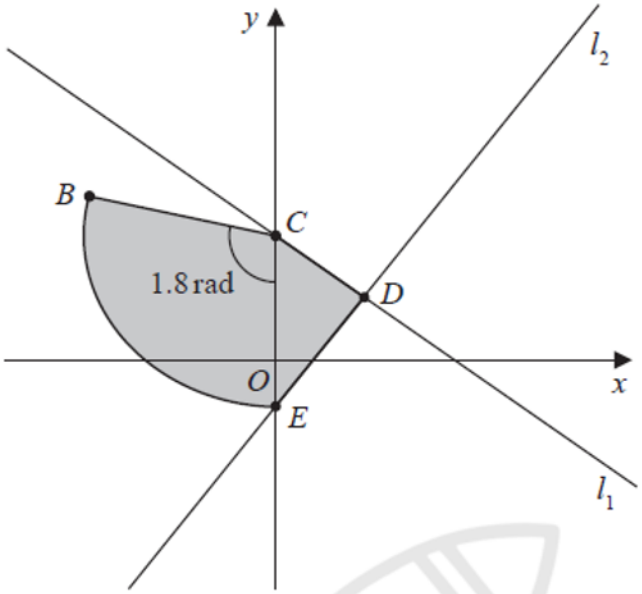


Figure 3

The line l_1 has equation $4y + 3x = 48$

The line l_1 cuts the y-axis at the point C, as shown in Figure 3.

(a) State the y coordinate of C.

(1)

The point D (8, 6) lies on l_1

The line l_2 passes through D and is perpendicular to l_1

The line l_2 cuts the y-axis at the point E as shown in Figure 3.

(b) Show that the y coordinate of E is $-\frac{14}{3}$

www.RocketRevise.com

(3)

A sector BCE of a circle with centre C is also shown in Figure 3.

Given that angle BCE is 1.8 radians,

(c) find the length of arc BE.

(3)

The region $CBED$, shown shaded in Figure 3, consists of the sector BCE joined to the triangle

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

CDE.

(d) Calculate the exact area of the region *CBED*.

(3)

(Total for question = 10 marks)

(Q07 WMA11/01, June 2021)

RocketRevise

Q17.

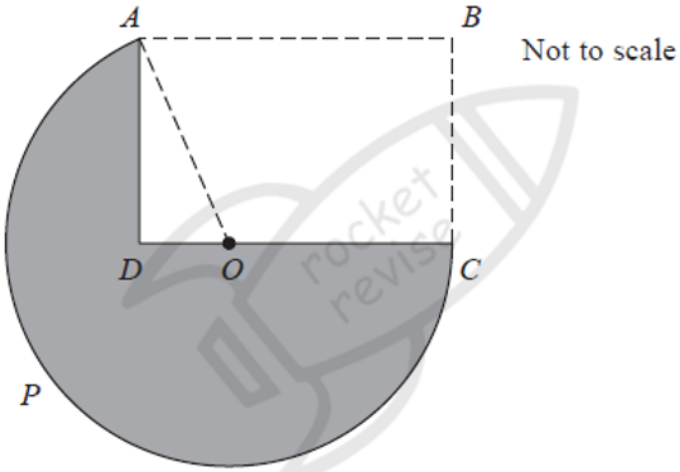


Figure 2

Figure 2 shows the plan view of a house *ABCD* and a lawn *APCDA*.

ABCD is a rectangle with *AB* = 16 m.

APCOA is a sector of a circle centre *O* with radius 12 m.

The point *O* lies on the line *DC*, as shown in Figure 2.

www.RocketRevise.com

(a) Show that the size of angle *AOD* is 1.231 radians to 3 decimal places.

(2)

The lawn *APCDA* is shown shaded in Figure 2.

(b) Find the area of the lawn, in m^2, to one decimal place.

(4)

(c) Find the perimeter of the lawn, in metres, to one decimal place.

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(3)

(Total for question = 9 marks)

(Q04 WMA11/01, Jan 2020)

Q18.

RocketRevise

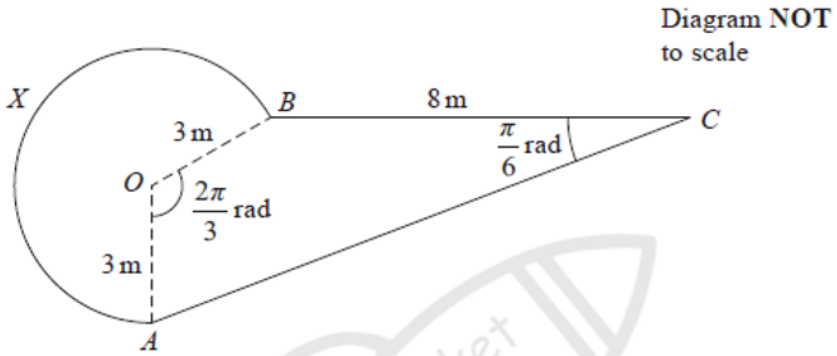


Figure 2

Figure 2 shows the plan view of a design for a pond.

The design consists of a sector *AOBX* of a circle centre *O* joined to a quadrilateral *AOBC*.

- $BC = 8\,\text{m}$
- $OA = OB = 3\,\text{m}$
- angle AOB is $\frac{2\pi}{3}$ radians
- angle BCA is $\frac{\pi}{6}$ radians

(a) Calculate (i) the exact area of the sector *AOBX*,

(ii) the exact perimeter of the sector *AOBX*.

www.RocketRevise.com

(5)

(b) Calculate the exact area of the triangle *AOB*.

(2)

(c) Show that the length *AB* is $3\sqrt{3}\,\text{m}$.

(2)

(d) Find the total surface area of the pond. Give your answer in m^2 correct to 2 significant

www.RocketRevise.com

compiled by Ahmed Gul
www.RocketRevise.com

figures.

(5)

(Total for question = 14 marks)

(Q08 WMA11/01, Oct 2022)

RocketRevise

Q19.

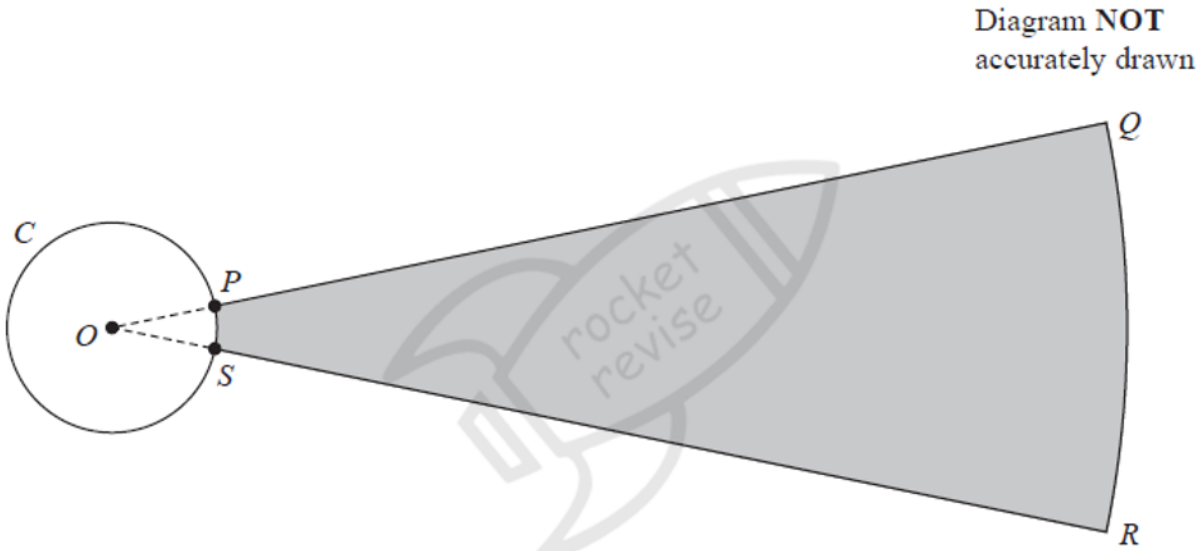


Figure 3

Figure 3 shows the plan view of the area being used for a ball-throwing competition.

Competitors must stand within the circle C and throw a ball as far as possible into the target area, $PQRS$, shown shaded in Figure 3.

Given that

- circle C has centre O
- P and S are points on C
- $OPQRSO$ is a sector of a circle with centre O
- the length of arc PS is 0.72 m
- the size of angle POS is 0.6 radians

www.RocketRevise.com

(a) show that $OP = 1.2$ m

(1)

Given also that

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

- the target area, *PQRS*, is 90 m^2
- length $PQ = x$ metres

(b) show that

$$5x^2 + 12x - 1500 = 0$$

(3)

(c) Hence calculate the total perimeter of the target area, *PQRS*, giving your answer to the nearest metre.

(3)

RocketRevise

(Total for question = 7 marks)

(Q09 WMA11/01, Oct 2023)

Q20.

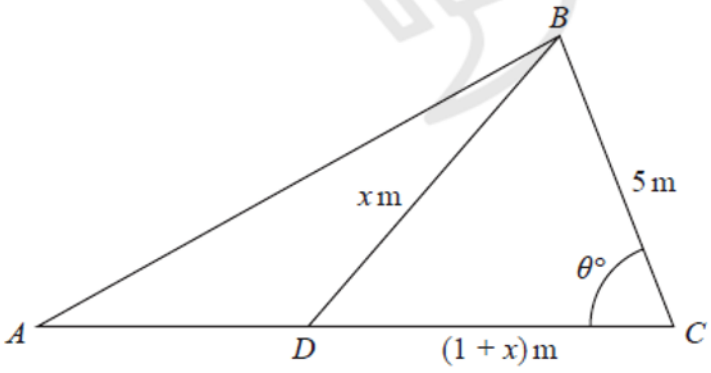


Diagram **NOT** accurately drawn

Figure 2

www.RocketRevise.com

Figure 2 shows the plan view of a frame for a flat roof.

The shape of the frame consists of triangle *ABD* joined to triangle *BCD*.

Given that

- $BD = x$ m
- $CD = (1 + x)$ m
- $BC = 5$ m
- angle $BCD = \theta°$

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(a) show that $\cos\theta° = \dfrac{13 + x}{5 + 5x}$

(2)

Given also that

- $x = 2\sqrt{3}$
- angle $BAC = 30°$
- ADC is a straight line

(b) find the area of triangle ABC, giving your answer, in m^2, to one decimal place.

RocketRevise

(5)

(Total for question = 7 marks)

(Q05 WMA11/01, Oct 2023)

www.RocketRevise.com

www.RocketRevise.com